

Sales & Profit Analysis Dashboard

Leveraging Power BI to transform raw transactional data into actionable business intelligence

POWER BI · DAX · POWER QUERY





PROJECT OVERVIEW

Why This Analysis Matters

Organisations sitting on transactional data without a structured analytical layer are leaving revenue on the table. This project builds an end-to-end Power BI dashboard that converts raw sales records into a clear picture of profitability, regional performance, and customer behaviour.

- ❗ Designed for decision-makers who need answers — not spreadsheets.

Project Objectives

Measure Performance

Track total sales, profit, and margins across categories, regions, and time periods.

Identify Loss Areas

Pinpoint sub-categories and regions where costs erode profit margins.

Analyse Payment Trends

Understand which payment modes drive volume and which correlate with higher order values.

Reveal Seasonality

Surface monthly and quarterly sales trends to support demand planning.



Dataset Overview

The dataset captures order-level transactional records across a multi-category retail operation. Key fields include:

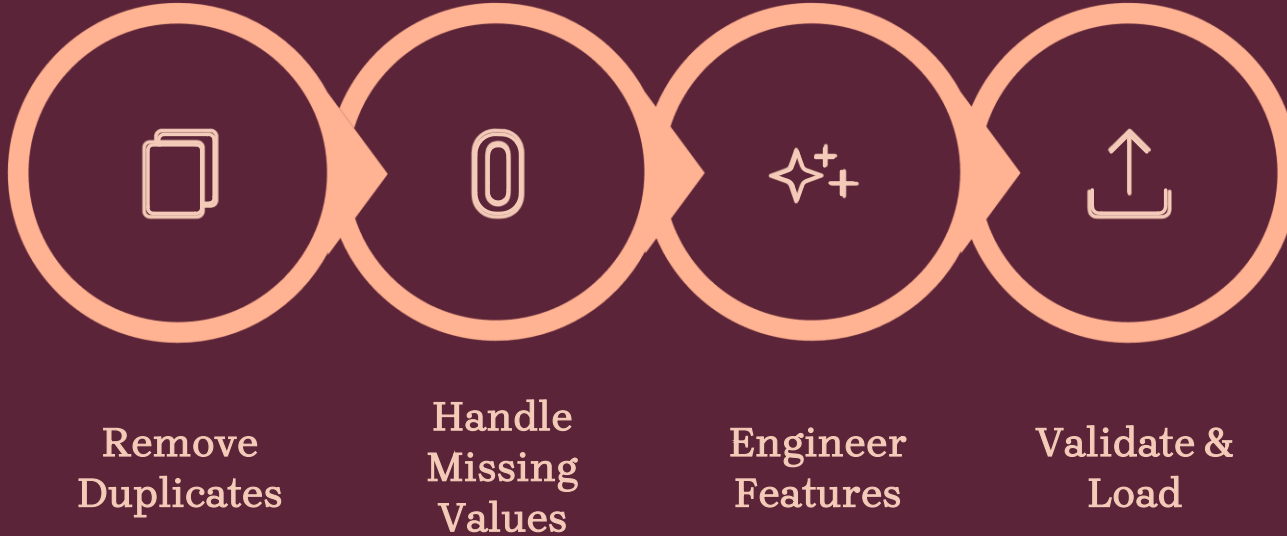
Field	Description
Order ID	Unique identifier per transaction
Category / Sub-Category	Product classification hierarchy
Sales	Gross revenue per line item (\$)
Profit	Net profit after cost deduction (\$)
Quantity	Units sold per order
Payment Mode	Card, COD, Online, etc.
Region / State	Geographic dimension for mapping
Order Date	Date of purchase for trend analysis

**Dataset scale:** ~10,000 order records spanning multiple years, product categories, and US regions — sufficient for statistically meaningful insights.

The **Profit Margin** column was derived during preparation:

$$\text{Profit Margin} = \frac{\text{Profit}}{\text{Sales}} \times 100$$

Data Cleaning & Preparation



A clean, consistent data model is the foundation of trustworthy insights. Each step was executed inside **Power Query** before the data was loaded into the analytical layer.

Tools & Technologies



Power BI Desktop

Core visualisation environment for building interactive reports, slicers, and the final dashboard layout.



Power Query (M)

ETL layer used for ingesting, transforming, and shaping raw CSV data before it enters the model.



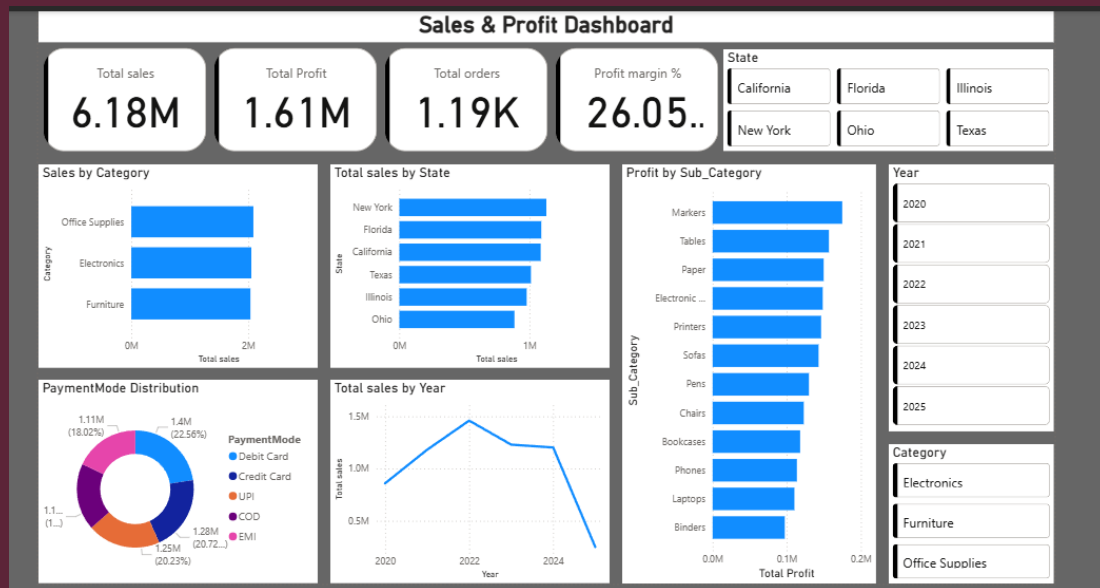
DAX

Custom measures including *Total Profit*, *Profit Margin %*, *YoY Growth*, and dynamic KPI cards.



What the Dashboard Contains

The dashboard is structured across **two report pages** — an executive summary and a drill-down detail view — with cross-page slicers for Region, Category, and Date Range.



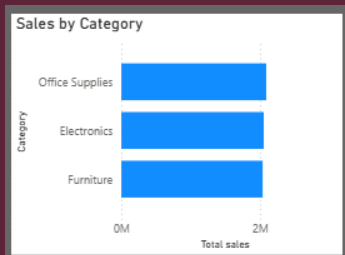
Page 1 — Executive Summary

KPI cards, sales trend, category breakdown, and regional map.

Page 2 — Profit Deep Dive

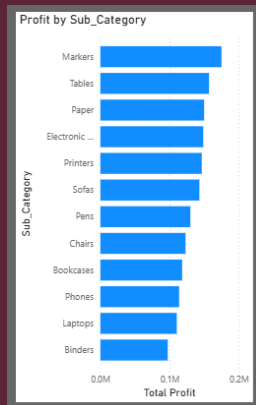
Sub-category profitability, payment mode analysis, and margin waterfall.

Key Visualisations



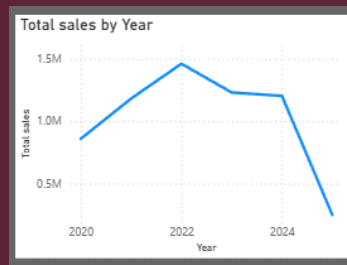
Sales by Category

Donut chart comparing Technology, Furniture, and Office Supplies — highlighting revenue concentration.



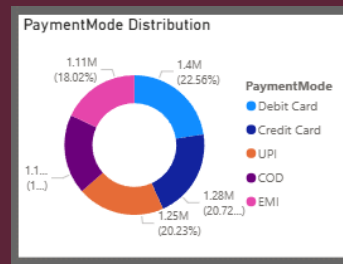
Profit by Sub-Category

Ranked bar chart exposing high-margin winners and loss-making sub-categories at a glance.



Monthly Sales Trend

Area line chart revealing seasonality spikes in Q4 and mid-year dips for demand planning.



Sales by Payment Mode

Stacked bar comparing COD, Credit Card, and Online payments — tied to average order value.

Key Business Insights

1

Technology Leads Revenue

Technology accounts for ~36% of total sales and delivers the highest absolute profit, driven by Phones and Accessories.

2

Furniture Is a Margin Trap

Tables and Bookcases consistently generate **negative profit** — high sales volume masks significant losses on these SKUs.

3

West Region Outperforms

The West contributes the highest profit margin regionally; the Central region underperforms despite moderate sales volume.

4

Q4 Seasonal Spike

Sales peak sharply in November–December, suggesting strong holiday demand that can be leveraged with targeted promotions.



Business Recommendations

Reassess Furniture Pricing

Conduct SKU-level cost analysis on Tables and Bookcases. Consider renegotiating supplier contracts or adjusting retail pricing to restore margin.

Invest in the West

Replicate the West region's sales model — promotional mix, channel strategy, and customer segmentation — in underperforming Central markets.

Capitalise on Q4 Demand

Pre-position inventory and launch targeted campaigns in October to maximise the proven Q4 revenue spike without margin compression.

Promote High-Value Payment Channels

Online and Credit Card transactions correlate with larger basket sizes. Incentivise these channels with loyalty points or instant discounts to lift average order value.

Double Down on Technology

Increase marketing spend on Phones and Accessories — the highest-margin sub-categories — while reviewing Office Supplies for rationalisation.